1)

Using “cp -r” //recursive copy

```
vignesh@DESKTOPGV:~$ mkdir linux_nw_mod1
vignesh@DESKTOPGV:~$ cd linux_nw_mod1/
vignesh@DESKTOPGV:~/linux_nw_mod1$ ls
vignesh@DESKTOPGV:~/linux_nw_mod1$ touch prog1.txt
vignesh@DESKTOPGV:~/linux_nw_mod1$ touch prog2.txt
vignesh@DESKTOPGV:~/linux_nw_mod1$ touch prog3.txt
vignesh@DESKTOPGV:~/linux_nw_mod1$ touch prog4.txt
vignesh@DESKTOPGV:~/linux_nw_mod1$ touch prog5.txt
vignesh@DESKTOPGV:~/linux_nw_mod1$ cd ..
vignesh@DESKTOPGV:~$ mkdir linux_nw_mod1test
vignesh@DESKTOPGV:~$ cp -r linux_nw_mod1 linux_nw_mod1test
vignesh@DESKTOPGV:~$ cd linux_nw_mod1test/
vignesh@DESKTOPGV:~/linux_nw_mod1test$ ls
linux_nw_mod1
vignesh@DESKTOPGV:~/linux_nw_mod1test$
```

USING SCP

```
rohit@EURLTP-429: /home
rohit@EURLTP-429:/home$ scp /home/module1/file1.txt rohit@172.17.116.116:/home/rohi
t/
rohit@172.17.116.116's password:
file1.txt                                     100%    0     0.0KB/s   00:00
rohit@EURLTP-429:/home$ sudo service ssh status
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-03-04 10:53:38 UTC; 19min ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 2844 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 2845 (sshd)
      Tasks: 1 (limit: 9320)
     Memory: 1.8M (peak: 20.0M)
        CPU: 197ms
     CGroup: /system.slice/ssh.service
             └─2845 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 04 10:53:38 EURLTP-429 systemd[1]: Started ssh.service - OpenBSD Secure Shell >
Mar 04 11:11:02 EURLTP-429 sshd[2981]: Accepted password for rohit from 172.17.116>
Mar 04 11:11:02 EURLTP-429 sshd[2981]: pam_unix(sshd:session): session opened for >
Mar 04 11:11:02 EURLTP-429 sshd[2981]: pam_unix(sshd:session): session closed for >
Mar 04 11:12:05 EURLTP-429 sshd[3024]: Accepted password for rohit from 172.17.116>
Mar 04 11:12:05 EURLTP-429 sshd[3024]: pam_unix(sshd:session): session opened for >
Mar 04 11:12:06 EURLTP-429 sshd[3024]: pam_unix(sshd:session): session closed for >
Mar 04 11:12:16 EURLTP-429 sshd[3067]: Accepted password for rohit from 172.17.116>
Mar 04 11:12:16 EURLTP-429 sshd[3067]: pam_unix(sshd:session): session opened for >
Mar 04 11:12:16 EURLTP-429 sshd[3067]: pam_unix(sshd:session): session closed for >
lines 1-24/24 (END)
```

2)

FTP

```
vignesh@DESKTOPGV:~$ sudo apt install vsftpd -y
vignesh@DESKTOPGV:~$ sudo systemctl start vsftpd
vignesh@DESKTOPGV:~$ sudo systemctl enable vsftpd
Synchronizing state of vsftpd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable vsftpd
vignesh@DESKTOPGV:~$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
     Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; prese>
     Active: active (running) since Tue 2026-03-03 14:47:39 UTC; 25s ago
   Main PID: 1194 (vsftpd)
      Tasks: 1 (limit: 9115)
     Memory: 700.0K ()
     CGroup: /system.slice/vsftpd.service
             └─1194 /usr/sbin/vsftpd /etc/vsftpd.conf

Mar 03 14:47:39 DESKTOPGV systemd[1]: Starting vsftpd.service - vsftpd FTP >
Mar 03 14:47:39 DESKTOPGV systemd[1]: Started vsftpd.service - vsftpd FTP s>
vignesh@DESKTOPGV:~$ sudo adduser rohit45
info: Adding user `rohit45' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `rohit45' (1001) ...
info: Adding new user `rohit45' (1001) with group `rohit45 (1001)' ...
info: Creating home directory `/home/rohit45' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for rohit45
Enter the new value, or press ENTER for the default
```

```
        Full Name []: Rohit45
        Room Number []: 45
        Work Phone []: 9865799783
        Home Phone []: 8248830309
        Other []:
Is the information correct? [Y/n] Y
info: Adding new user `rohit45' to supplemental / extra groups `users' ...
info: Adding user `rohit45' to group `users' ...
vignesh@DESKTOPGV:~$ sudo nano /etc/vsftpd.conf
vignesh@DESKTOPGV:~$ sudo systemctl restart vsftpd
vignesh@DESKTOPGV:~$ sudo nano /etc/vsftpd.conf
vignesh@DESKTOPGV:~$ sudo nano /etc/vsftpd.conf
vignesh@DESKTOPGV:~$ sudo systemctl restart vsftpd
vignesh@DESKTOPGV:~$ ip a
vignesh@DESKTOPGV:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1420
        inet 172.23.98.200  netmask 255.255.240.0  broadcast 172.23.111.255
        inet6 fe80::215:5dff:fe7e:565a   prefixlen 64  scopeid 0x20<link>
        ether 00:15:5d:7e:56:5a   txqueuelen 1000  (Ethernet)
        RX packets 11571  bytes 15953838 (15.9 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1531  bytes 120688 (120.6 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 54  bytes 6770 (6.7 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
```

```
        TX packets 54  bytes 6770 (6.7 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

vignesh@DESKTOPGV:~$ sudo nano /etc/vsftpd.conf
vignesh@DESKTOPGV:~$ sudo systemctl restart vsftpd
vignesh@DESKTOPGV:~$ ftp localhost
Connected to localhost.
220 (vsFTPd 3.0.5)
Name (localhost:vignesh):  rohit45
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> put test.c
200 PORT command successful. Consider using PASV.
150 Ok to send data.
226 Transfer complete.
403 bytes sent in 0.00444 seconds (88.7 kbytes/s)
ftp> get test.c
200 PORT command successful. Consider using PASV.
150 Opening BINARY mode data connection for test.c (403 bytes).
226 Transfer complete.
403 bytes received in 0.000326 seconds (1.18 Mbytes/s)
ftp> ^Z
[2]+  Stopped                 ftp localhost
vignesh@DESKTOPGV:~$
```

SFTP

```
vignesh@DESKTOPGV:~$ sudo apt install openssh-server -y
vignesh@DESKTOPGV:~$ sudo systemctl status ssh
```

○ ssh.service - OpenBSD Secure Shell server

Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset:>

Active: inactive (dead)

TriggeredBy: ● ssh.socket

Docs: man:sshd(8)

man:sshd_config(5)

vignesh@DESKTOPGV:~$ sudo systemctl start ssh

vignesh@DESKTOPGV:~$ sudo systemctl enable ssh

Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.

Executing: /usr/lib/systemd/systemd-sysv-install enable ssh

Created symlink /etc/systemd/system/sshd.service → /usr/lib/systemd/system/ssh.service.

Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service.

vignesh@DESKTOPGV:~$ hostname _I

hostname: the specified hostname is invalid

vignesh@DESKTOPGV:~$ hostname -I

172.23.98.200

vignesh@DESKTOPGV:~$ sftp rohit45@localhost

The authenticity of host 'localhost (127.0.0.1)' can't be established.

ED25519 key fingerprint is SHA256:mBxWkgjttVy6PBRSBo8kKoFEV2AEhGQY5/E0fnFxFY0.

This key is not known by any other names.

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes

Warning: Permanently added 'localhost' (ED25519) to the list of known hosts.

rohit45@localhost's password:

Connected to localhost.

sftp> ls

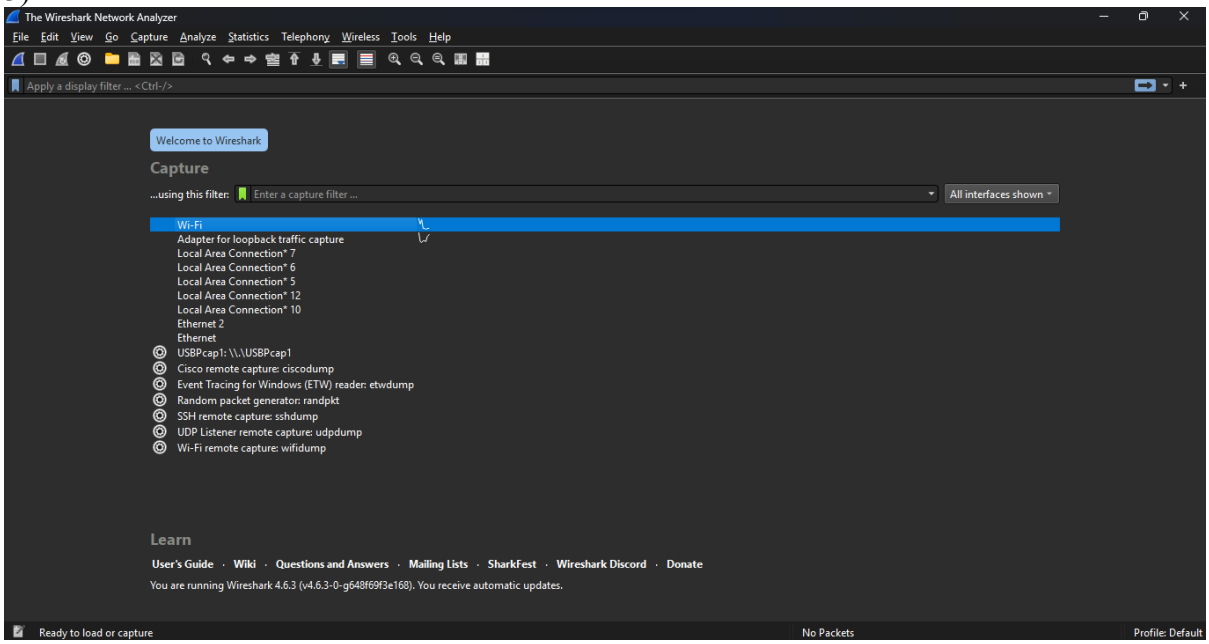
test.c

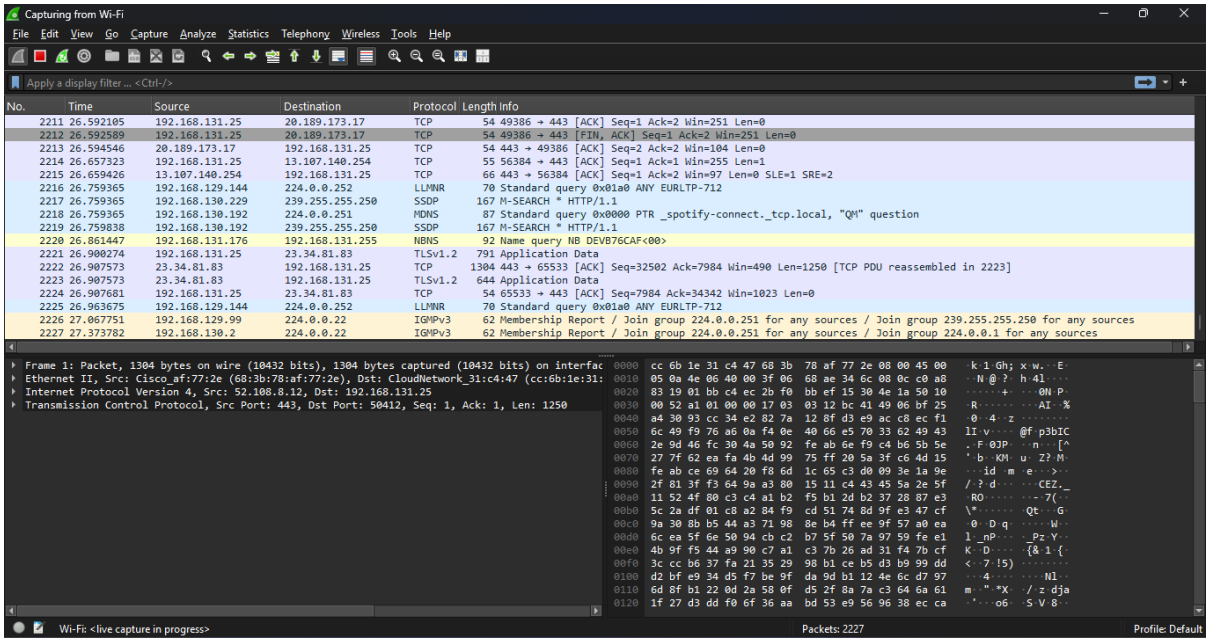
sftp> pwd

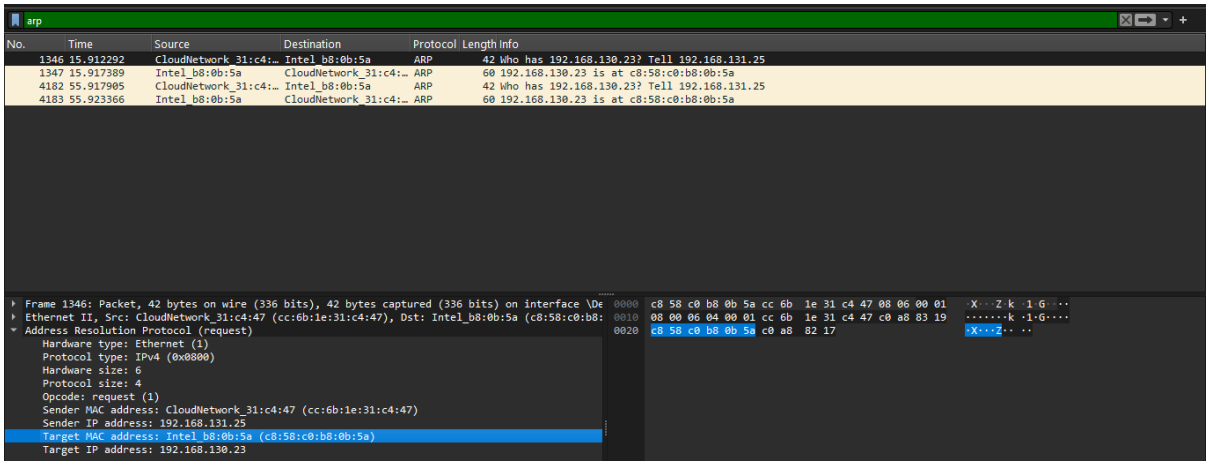
Remote working directory: /home/rohit45

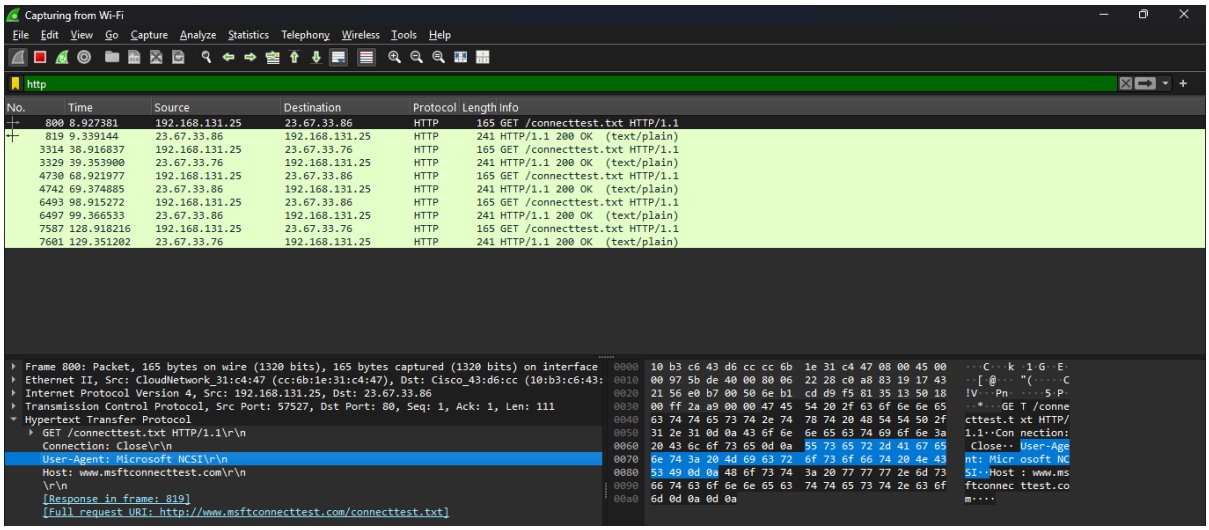
```
sftp> get test.c
Fetching /home/rohit45/test.c  to test.c
test.c                                100%  403   111.5KB/s   00:00
sftp> put test.c
Uploading test.c to /home/rohit45/test.c
test.c                                100%  403   336.4KB/s   00:00
sftp>
[3]+  Stopped                 sftp rohit45@localhost
```

3) WIRESHARK









4) LINUX UTILITIES PING

```
rohit@EURLTP-429: ~
rohit@EURLTP-429:~$ ping google.com
PING google.com (142.251.32.46) 56(84) bytes of data.
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=1 ttl=113 time=220 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=2 ttl=113 time=225 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=3 ttl=113 time=239 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=4 ttl=113 time=224 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=5 ttl=113 time=220 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=6 ttl=113 time=219 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=7 ttl=113 time=218 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=8 ttl=113 time=235 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=9 ttl=113 time=219 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=10 ttl=113 time=222 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=11 ttl=113 time=220 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=12 ttl=113 time=220 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=13 ttl=113 time=223 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=14 ttl=113 time=232 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=15 ttl=113 time=217 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=16 ttl=113 time=243 ms
64 bytes from sfo03s26-in-f14.1e100.net (142.251.32.46): icmp_seq=17 ttl=113 time=219 ms
```

```
rohit@EURLTP-429: ~
rohit@EURLTP-429:~$ sudo tcpdump -i eth0
[sudo] password for rohit:
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
07:28:58.396353 IP 172.17.116.116.54630 > alphyn.canonical.com.ntp: NTPv4, Client, length 48
07:28:58.614517 IP alphyn.canonical.com.ntp > 172.17.116.116.54630: NTPv4, Server, length 48
07:29:16.025696 IP6 fe80::215:5dff:fed3:c9d > ip6-allrouters: ICMP6, router solicitation, length 16
07:29:19.585900 IP 172.17.116.116 > sfo03s26-in-f14.1e100.net: ICMP echo request, id 958, seq 1, length 64
07:29:19.805643 IP sfo03s26-in-f14.1e100.net > 172.17.116.116: ICMP echo reply, id 958, seq 1, length 64
07:29:19.920526 IP6 fe80::c3de:b6ce:3866:7fd5.dhcpv6-client > ff02::1:2.dhcpv6-server: dhcp6 inf-req
07:29:20.590790 IP 172.17.116.116 > sfo03s26-in-f14.1e100.net: ICMP echo request, id 958, seq 2, length 64
07:29:20.815635 IP sfo03s26-in-f14.1e100.net > 172.17.116.116: ICMP echo reply, id 958, seq 2, length 64
07:29:21.602995 IP 172.17.116.116 > sfo03s26-in-f14.1e100.net: ICMP echo request, id 958, seq 3, length 64
07:29:21.842127 IP sfo03s26-in-f14.1e100.net > 172.17.116.116: ICMP echo reply, id 958, seq 3, length 64
07:29:22.615487 IP 172.17.116.116 > sfo03s26-in-f14.1e100.net: ICMP echo request, id 958, seq 4, length 64
07:29:22.839891 IP sfo03s26-in-f14.1e100.net > 172.17.116.116: ICMP echo reply, id 958, seq 4, length 64
07:29:23.626308 IP 172.17.116.116 > sfo03s26-in-f14.1e100.net: ICMP echo request, id 958, seq 5, length 64
07:29:30.646219 IP 172.17.116.116 > sfo03s26-in-f14.1e100.net: ICMP echo request, id 958, seq 12, length 64
07:29:30.681681 IP 172.17.116.116.52637 > alphyn.canonical.com.ntp: NTPv4, Client, length 48
07:29:30.865890 IP sfo03s26-in-f14.1e100.net > 172.17.116.116: ICMP echo reply, id 958, seq 12, length 64
07:29:30.910409 IP alphyn.canonical.com.ntp > 172.17.116.116.52637: NTPv4, Server, length 48
```

```
rohit@EURLTP-429: ~
rohit@EURLTP-429:~$ ping -c 2 embedur.ai
PING embedur.ai (162.159.137.54) 56(84) bytes of data.
64 bytes from 162.159.137.54: icmp_seq=1 ttl=58 time=5.51 ms
64 bytes from 162.159.137.54: icmp_seq=2 ttl=58 time=5.86 ms

--- embedur.ai ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 999ms
rtt min/avg/max/mdev = 5.509/5.682/5.855/0.173 ms
rohit@EURLTP-429:~$ ping -i 5 embedur.ai
PING embedur.ai (162.159.136.54) 56(84) bytes of data.
64 bytes from 162.159.136.54: icmp_seq=1 ttl=57 time=21.7 ms
64 bytes from 162.159.136.54: icmp_seq=2 ttl=57 time=5.54 ms
64 bytes from 162.159.136.54: icmp_seq=3 ttl=57 time=5.63 ms
^C
--- embedur.ai ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 10093ms
rtt min/avg/max/mdev = 5.539/10.946/21.669/7.582 ms
rohit@EURLTP-429:~$ ping -s 47 -c 2 embedur.ai
PING embedur.ai (162.159.137.54) 47(75) bytes of data.
55 bytes from 162.159.137.54: icmp_seq=1 ttl=57 time=6.33 ms
55 bytes from 162.159.137.54: icmp_seq=2 ttl=57 time=6.06 ms

--- embedur.ai ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 999ms
rtt min/avg/max/mdev = 6.060/6.196/6.332/0.136 ms
rohit@EURLTP-429:~$ ping -W 5 -c 3 embedur.ai
PING embedur.ai (162.159.136.54) 56(84) bytes of data.
64 bytes from 162.159.136.54: icmp_seq=1 ttl=58 time=6.01 ms
64 bytes from 162.159.136.54: icmp_seq=2 ttl=58 time=5.47 ms
64 bytes from 162.159.136.54: icmp_seq=3 ttl=58 time=6.56 ms

--- embedur.ai ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2000ms
rtt min/avg/max/mdev = 5.471/6.014/6.562/0.445 ms
rohit@EURLTP-429:~$ ping -w 1 -c 3 embedur.ai
PING embedur.ai (162.159.137.54) 56(84) bytes of data.
64 bytes from 162.159.137.54: icmp_seq=1 ttl=57 time=6.15 ms

--- embedur.ai ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 6.150/6.150/6.150/0.000 ms
rohit@EURLTP-429:~$ ping -w 1 -c 3 -6 embedur.ai
ping: embedur.ai: Address family for hostname not supported
```

```
rohit@EURLTP-429:~$ ping -w 1 -c 3 -4 embedur.ai
PING embedur.ai (162.159.137.54) 56(84) bytes of data.
64 bytes from 162.159.137.54: icmp_seq=1 ttl=57 time=6.16 ms

--- embedur.ai ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 6.162/6.162/6.162/0.000 ms
rohit@EURLTP-429:~$ ping -I eth0 -c 3  embedur.ai
PING embedur.ai (162.159.136.54) from 172.17.116.116 eth0: 56(84) bytes of data.
64 bytes from 162.159.136.54: icmp_seq=1 ttl=58 time=4.15 ms
64 bytes from 162.159.136.54: icmp_seq=2 ttl=58 time=5.57 ms
64 bytes from 162.159.136.54: icmp_seq=3 ttl=58 time=6.07 ms

--- embedur.ai ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 1998ms
rtt min/avg/max/mdev = 4.148/5.259/6.066/0.812 ms
rohit@EURLTP-429:~$ ping -D -c 3  embedur.ai
PING embedur.ai (162.159.137.54) 56(84) bytes of data.
[1772609983.042348] 64 bytes from 162.159.137.54: icmp_seq=1 ttl=58 time=4.13 ms
[1772609983.813327] 64 bytes from 162.159.137.54: icmp_seq=2 ttl=58 time=5.72 ms
[1772609984.815044] 64 bytes from 162.159.137.54: icmp_seq=3 ttl=58 time=4.70 ms

--- embedur.ai ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 1998ms
rtt min/avg/max/mdev = 4.134/4.853/5.721/0.656 ms
rohit@EURLTP-429:~$
```

Using ARP

```
rohit@EURLTP-429: ~
ohit@EURLTP-429:~$ arp -a
URLTP-429.mshome.net (172.17.112.1) at 00:15:5d:90:db:c0 [ether] on eth0
ohit@EURLTP-429:~$ arp -n
ddress                  HWtype  HWaddress           Flags Mask            Iface
72.17.112.1             ether   00:15:5d:90:db:c0   C                     eth0
ohit@EURLTP-429:~$ arp -v
ddress                  HWtype  HWaddress           Flags Mask            Iface
URLTP-429.mshome.net    ether   00:15:5d:90:db:c0   C                     eth0
ntries: 1      Skipped: 0      Found: 1
ohit@EURLTP-429:~$ arp -i lo
rp: in 1 entries no match found.
ohit@EURLTP-429:~$ arp -i eth0
ddress                  HWtype  HWaddress           Flags Mask            Iface
URLTP-429.mshome.net    ether   00:15:5d:90:db:c0   C                     eth0
ohit@EURLTP-429:~$ arp -s 192.168.1.10 00:11:22:33:44:55:
IOCSARP: Operation not permitted
ohit@EURLTP-429:~$ sudo arp -s 192.168.1.10 00:11:22:33:44:55
sudo] password for rohit:
IOCSARP: Network is unreachable
ohit@EURLTP-429:~$ sudo arp -s 172.17.112.50 00:11:22:33:44:55
ohit@EURLTP-429:~$ arp -a
 (172.17.112.50) at 00:11:22:33:44:55 [ether] PERM on eth0
URLTP-429.mshome.net (172.17.112.1) at 00:15:5d:90:db:c0 [ether] on eth0
ohit@EURLTP-429:~$ sudo arp -d 172.17.112.50
ohit@EURLTP-429:~$ arp -a
URLTP-429.mshome.net (172.17.112.1) at 00:15:5d:90:db:c0 [ether] on eth0
ohit@EURLTP-429:~$
```

5)

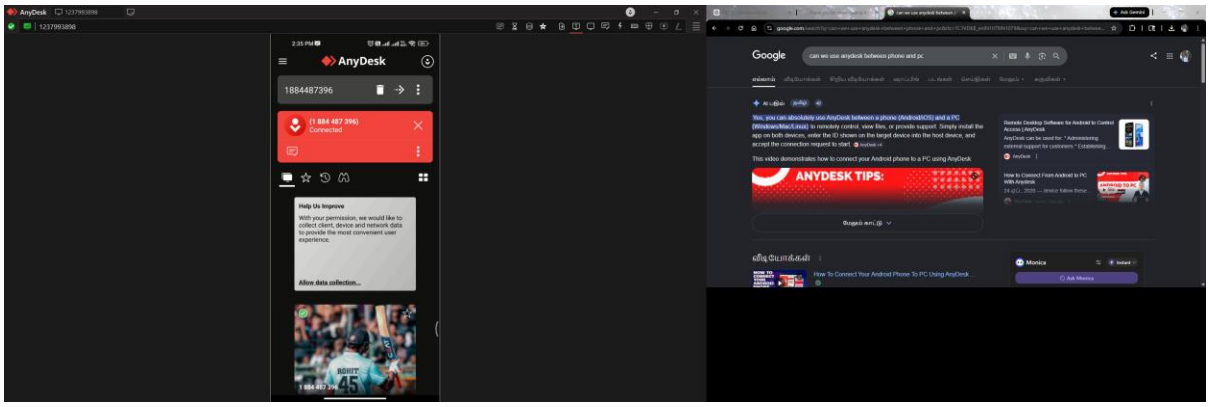
A duplicate IP occurs when two devices on the same network are assigned the same IP address. This can happen due to:

- Manual static IP assignment without coordination
- DHCP misconfiguration (assigning the same IP to multiple clients)
- Network mismanagement or cloning virtual machines with identical IPs

When two devices on the same network are assigned the same IP address, it creates an IP conflict, which can cause serious network problems. Both devices will attempt to use the

same address to communicate, leading to confusion in the network. Because network communication relies on each IP being unique, packets intended for one device may be received by the other, causing intermittent connectivity or complete loss of communication. Operating systems and network devices detect the conflict in different ways. Typically, they use the Address Resolution Protocol (ARP) to map IP addresses to physical MAC addresses. When two devices claim the same IP, ARP tables on switches, routers, and computers become inconsistent, resulting in flapping MAC addresses and incorrect routing of packets. This can cause applications and services to fail, active TCP connections to drop, and network resources such as printers or shared folders to become unreachable. Some operating systems will display an error message indicating a duplicate IP, such as "IP address conflict detected" on Windows or "Duplicate address detected" on Linux and Unix systems. DHCP servers may detect the conflict if one or both devices use dynamically assigned IPs and may attempt to resolve it by refusing to lease the duplicate address, which can result in devices repeatedly requesting new IPs. The overall impact on the network can include instability, packet loss, routing errors, intermittent service failures, and potential security risks, because traffic may be delivered to the wrong device. To resolve such conflicts, an administrator must reassign one of the conflicting IP addresses or remove static assignments that overlap with the DHCP pool. Best practices to avoid duplicate IPs include using DHCP for dynamic assignments, maintaining a centralized IP address plan for static addresses, monitoring ARP tables or using network management tools to detect conflicts, and avoiding manual assignment of addresses that fall within the DHCP range. In summary, duplicate IP addresses disrupt normal communication, cause network instability, and require administrative intervention to restore proper operation.

6)



7)

```
Select rohit@EURLTP-429: ~
rohit@EURLTP-429:~$ ip route
default via 172.17.112.1 dev eth0 proto kernel
172.17.112.0/20 dev eth0 proto kernel scope link src 172.17.116.116
172.18.0.0/16 dev docker0 proto kernel scope link src 172.18.0.1 linkdown
rohit@EURLTP-429:~$ ping 172.17.112.1
PING 172.17.112.1 (172.17.112.1) 56(84) bytes of data.
64 bytes from 172.17.112.1: icmp_seq=1 ttl=128 time=0.394 ms
64 bytes from 172.17.112.1: icmp_seq=2 ttl=128 time=0.368 ms
64 bytes from 172.17.112.1: icmp_seq=3 ttl=128 time=0.830 ms
64 bytes from 172.17.112.1: icmp_seq=4 ttl=128 time=0.828 ms
^C
--- 172.17.112.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3062ms
rtt min/avg/max/mdev = 0.368/0.605/0.830/0.224 ms
rohit@EURLTP-429:~$ ip neigh
172.17.112.1 dev eth0 lladdr 00:15:5d:90:db:c0 REACHABLE
rohit@EURLTP-429:~$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  EURLTP-429.mshome.net (172.17.112.1)  0.390 ms  0.319 ms  0.301 ms
 2  192.168.128.1 (192.168.128.1)  3.245 ms  6.535 ms  7.386 ms
 3  192.168.139.67 (192.168.139.67)  7.324 ms  3.890 ms  5.661 ms
 4  182.79.20.101 (182.79.20.101)  11.467 ms static-201.140.16.61-tataidc.co.in (61
.16.140.201)  59.674 ms  59.550 ms
 5  142.250.171.162 (142.250.171.162)  7.332 ms *  7.250 ms
 6  * * *
 7  * dns.google (8.8.8.8)  6.331 ms *
rohit@EURLTP-429:~$
```

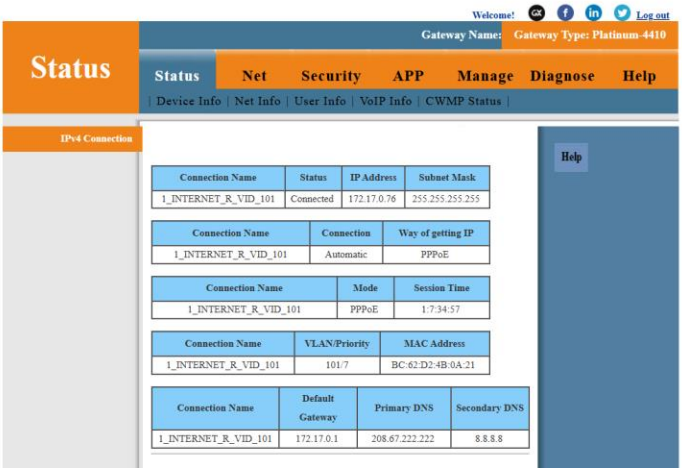
8)

```
rohit@EURLTP-429: ~
rohit@EURLTP-429:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 172.17.116.116  netmask 255.255.240.0  broadcast 172.17.127.255
        inet6 fe80::215:5dff:fed3:c9d  prefixlen 64  scopeid 0x20<link>
        ether 00:15:5d:d3:0c:9d  txqueuelen 1000  (Ethernet)
        RX packets 14195  bytes 16294127 (16.2 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 3342  bytes 267761 (267.7 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 30701  bytes 2770891 (2.7 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 30701  bytes 2770891 (2.7 MB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

rohit@EURLTP-429:~$
```

9)



10)DHCP PROTOCOL

DHCP (Dynamic Host Configuration Protocol) is a network protocol that automatically assigns IP addresses and related network configuration parameters to devices on a network. This allows devices to communicate without manual configuration.

The standard DHCP process is known as DORA, which stands for Discover, Offer, Request, Acknowledge.

Step 1: DHCP Discover

- When a client connects to the network, it does not have an IP address.
- The client broadcasts a DHCP Discover message on the local network to find available DHCP servers.

Step 2: DHCP Offer

- The DHCP server receives the Discover message and selects an available IP address from its pool.
- It responds with a DHCP Offer, which includes:
    - Offered IP address
    - Subnet mask
    - Default gateway
    - DNS server information
    - Lease time

Step 3: DHCP Request

- The client receives one or more offers and selects one.
- It broadcasts a DHCP Request message to indicate which offered IP it intends to use.

Step 4: DHCP Acknowledgment (ACK)

- The DHCP server confirms the assignment with a DHCP ACK message.
- The server marks the IP as leased and provides the client with all configuration details.
- The client configures its network interface with the assigned IP address and other parameters.

11)

```
rohit@EURLTP-429:~$ sudo apt install openssh-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:9.6p1-3ubuntu13.14).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
rohit@EURLTP-429:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/sshd.service → /usr/lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service.
rohit@EURLTP-429:~$ sudo systemctl start ssh
rohit@EURLTP-429:~$ sudo service ssh status
● ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-03-04 09:51:56 UTC; 10s ago
TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 1875 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1876 (sshd)
```

```
     Tasks: 1 (limit: 9320)
    Memory: 2.2M (peak: 2.8M)
       CPU: 23ms
    CGroup: /system.slice/ssh.service
            └─1876 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Mar 04 09:51:56 EURLTP-429 systemd[1]: Starting ssh.service - OpenBSD Secure Shell>
Mar 04 09:51:56 EURLTP-429 sshd[1876]: Server listening on 0.0.0.0 port 22.
Mar 04 09:51:56 EURLTP-429 sshd[1876]: Server listening on :: port 22.
Mar 04 09:51:56 EURLTP-429 systemd[1]: Started ssh.service - OpenBSD Secure Shell >
rohit@EURLTP-429:~$ whoami
rohit
rohit@EURLTP-429:~$ hostname -I
172.17.116.116
rohit@EURLTP-429:~$ ssh rohit@172.17.116.116
The authenticity of host '172.17.116.116 (172.17.116.116)' can't be established.
ED25519 key fingerprint is
SHA256:sMKv+E4wOnJrPJaQIAtubdgTeLfUMrQLnpiY8i9AS1o.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.17.116.116' (ED25519) to the list of known hosts.
rohit@172.17.116.116's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

* Documentation:  https://help.ubuntu.com
* Management:     https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Wed Mar 4 09:53:48 UTC 2026

  System load:  0.11              Processes:           49
```

Usage of /: 0.5% of 1006.85GB Users logged in: 1

Memory usage: 5% IPv4 address for eth0: 172.17.116.116

Swap usage: 0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s

just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

rohit@EURLTP-429:~$

```
rohit@EURLTP-429: ~/hi
(c) Microsoft Corporation. All rights reserved.

C:\Users\user temp>ssh rohit@172.17.117.117

C:\Users\user temp>^Z

C:\Users\user temp>ssh rohit@172.17.116.116
The authenticity of host '172.17.116.116 (172.17.116.116)' can't be established.
ED25519 key fingerprint is SHA256:sMKv+E4wOnJrPJaQIAtubdgTeLfUMrQLnpiY8i9AS1o.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? tes
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '172.17.116.116' (ED25519) to the list of known hosts.
rohit@172.17.116.116's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Mar  4 10:02:30 UTC 2026

  System load:  0.0                Processes:             54
  Usage of /:   0.5% of 1006.85GB  Users logged in:       1
  Memory usage: 5%                 IPv4 address for eth0: 172.17.116.116
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Last login: Wed Mar  4 09:53:48 2026 from 172.17.116.116
rohit@EURLTP-429:~$ ls-l
ls-l: command not found
rohit@EURLTP-429:~$ ls -l
total 8
-rwxr-xr-x 1 rohit rohit 1426 Mar  3 11:14 installer.sh
drwxr-xr-x 2 rohit rohit 4096 Mar  3 11:12 nwmod1test
rohit@EURLTP-429:~$ ls
installer.sh  nwmod1test
rohit@EURLTP-429:~$ mkdir hi
rohit@EURLTP-429:~$ cd hi
rohit@EURLTP-429:~/hi$
```

```
rohit@EURLTP-429: ~
rohit@EURLTP-429:~$ ls
hi  installer.sh  nwmod1test
rohit@EURLTP-429:~$
```

```
 ESCOD
o ssh.service - OpenBSD Secure Shell server
     Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
     Active: inactive (dead) since Wed 2026-03-04 10:05:26 UTC; 1min 13s ago
   Duration: 13min 29.828s
TriggeredBy: o ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 1875 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Process: 1876 ExecStart=/usr/sbin/sshd -D $SSHD_OPTS (code=exited, status=0/SUCCESS)
   Main PID: 1876 (code=exited, status=0/SUCCESS)
        CPU: 163ms

Mar 04 09:51:56 EURLTP-429 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Mar 04 09:53:48 EURLTP-429 sshd[1888]: Accepted password for rohit from 172.17.116.116 port 60858 ssh2
Mar 04 09:53:48 EURLTP-429 sshd[1888]: pam_unix(sshd:session): session opened for user rohit(uid=1000) by rohit(uid=0)
Mar 04 09:58:12 EURLTP-429 sshd[1948]: Connection reset by 172.17.112.1 port 60075 [preauth]
Mar 04 10:02:30 EURLTP-429 sshd[1955]: Accepted password for rohit from 172.17.112.1 port 62662 ssh2
Mar 04 10:02:30 EURLTP-429 sshd[1955]: pam_unix(sshd:session): session opened for user rohit(uid=1000) by rohit(uid=0)
Mar 04 10:05:26 EURLTP-429 systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
Mar 04 10:05:26 EURLTP-429 sshd[1876]: Received signal 15; terminating.
Mar 04 10:05:26 EURLTP-429 systemd[1]: ssh.service: Deactivated successfully.
Mar 04 10:05:26 EURLTP-429 systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
```

rohit@EURLTP-429:~$ sudo apt install telnetd -y

Mar 04 10:11:42 EURLTP-429 systemd[1]: Starting inetutils-inetd.service - GNU Netw>

```
Mar 04 10:11:42 EURLTP-429 systemd[1]: inetutils-inetd.service: Skipped due to 'ex>
Mar 04 10:11:42 EURLTP-429 systemd[1]: Condition check resulted in inetutils-inetd>
rohit@EURLTP-429:~$ sudo nano /etc/inetd.conf
rohit@EURLTP-429:~$ sudo service inetutils-inetd start
rohit@EURLTP-429:~$ sudo service inetutils-inetd status
● inetutils-inetd.service - GNU Network Utilities internet superserver
     Loaded: loaded (/usr/lib/systemd/system/inetutils-inetd.service; enabled; pre>
     Active: active (running) since Wed 2026-03-04 10:12:54 UTC; 2s ago
       Docs: man:inetutils-inetd(8)
             https://www.gnu.org/software/inetutils/manual/
    Process: 2472 ExecCondition=grep -qr ^[0-9A-Za-z/] /etc/inetd.conf /etc/inetd.>
   Main PID: 2474 (inetutils-inetd)
      Tasks: 1 (limit: 9320)
     Memory: 296.0K (peak: 1.8M)
        CPU: 7ms
     CGroup: /system.slice/inetutils-inetd.service
             └─2474 /usr/sbin/inetutils-inetd --foreground

Mar 04 10:12:54 EURLTP-429 systemd[1]: Starting inetutils-inetd.service - GNU Netw>
Mar 04 10:12:54 EURLTP-429 systemd[1]: Started inetutils-inetd.service - GNU Netwo>
...skipping...
● inetutils-inetd.service - GNU Network Utilities internet superserver
     Loaded: loaded (/usr/lib/systemd/system/inetutils-inetd.service; enabled; pre>
     Active: active (running) since Wed 2026-03-04 10:12:54 UTC; 2s ago
       Docs: man:inetutils-inetd(8)
             https://www.gnu.org/software/inetutils/manual/
    Process: 2472 ExecCondition=grep -qr ^[0-9A-Za-z/] /etc/inetd.conf /etc/inetd.>
   Main PID: 2474 (inetutils-inetd)
      Tasks: 1 (limit: 9320)
     Memory: 296.0K (peak: 1.8M)
```

```
      CPU: 7ms
    CGroup: /system.slice/inetutils-inetd.service
            └─2474 /usr/sbin/inetutils-inetd  --foreground

Mar 04 10:12:54 EURLTP-429 systemd[1]: Starting inetutils-inetd.service  - GNU Netw>
Mar 04 10:12:54 EURLTP-429 systemd[1]: Started inetutils-inetd.service  - GNU Netwo>
rohit@EURLTP-429:~$ telnet 172.17.116.116
Trying 172.17.116.116...
Connected to 172.17.116.116.
Escape character is '^]'.

Linux 6.6.87.2-microsoft-standard-WSL2  (EURLTP-429.localdomain) (pts/7)

EURLTP-429 login:
Password: Connection closed by foreign host.
rohit@EURLTP-429:~$ telnet 172.17.116.116
Trying 172.17.116.116...
Connected to 172.17.116.116.
Escape character is '^]'.

Linux 6.6.87.2-microsoft-standard-WSL2  (EURLTP-429.localdomain) (pts/7)

EURLTP-429 login: rohit
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2  x86_64)

* Documentation:  https://help.ubuntu.com
* Management:     https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

System information as of Wed Mar 4 10:13:18 UTC 2026

System load: 0.0 Processes: 58

Usage of /: 0.5% of 1006.85GB Users logged in: 1

Memory usage: 6% IPv4 address for eth0: 172.17.116.116

Swap usage: 0%

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s just raised the bar for easy, resilient and secure K8s cluster deployment.

https://ubuntu.com/engage/secure-kubernetes-at-the-edge

rohit@EURLTP-429:~$

```
Telnet 172.17.116.116

Linux 6.6.87.2-microsoft-standard-WSL2 (EURLTP-429.localdomain) (pts/8)

EURLTP-429 login: rohit

Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Mar  4 10:31:12 UTC 2026

  System load:  0.0                 Processes:             61
  Usage of /:   0.5% of 1006.85GB   Users logged in:       1
  Memory usage: 6%                  IPv4 address for eth0: 172.17.116.116
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
rohit@EURLTP-429:~$ ls
hi  installer.sh  nwmod1test
rohit@EURLTP-429:~$
rohit@EURLTP-429:~$
```